RJ Lopez

FULL-STACK DEVELOPER · AI SYSTEMS BUILDER

VrtxOmega@pm.me
github.com/VrtxOmega
Bethalto, IL · Remote
Authorized to work in the US

Self-taught full-stack developer with **33 shipped applications** built in 12 weeks. I design and build complete systems — desktop apps, web platforms, AI integrations, and mobile — using **Python, Electron, React Native, Flask**, and cloud deployment. My background is unconventional: field service, cleanroom certification, restaurant management. That history built the instincts I code with — root cause analysis, systems thinking, working fast under pressure with no margin for error. **I don't wait for permission to build.**

33	431	12	0%
APPS SHIPPED	TEST SPECS	WEEKS ACTIVE	ERROR RATE

▸ FLAGSHIP SYSTEMS

VERITAS Core
Live AI verification oracle on Cloud Run — Gemini, Stripe billing, HuggingFace dataset, Zenodo DOI

▸ Published Research Artifact

Omega Brain MCP
Python MCP server for AI agent provenance — S.E.A.L. hash chain, 10-gate deterministic pipeline

▸ 82 Test Suites Passing

AEGIS Protect
System security & hardware health platform — direct-to-metal telemetry, automated remediation

▸ 73,762 Adversarial Events / 0% Failure

OmegaWallet
Cryptographic ledger system for autonomous agent transactions — immutable state, signed operations

▸ 356 Local Test Suites

Sovereign Media
AIO desktop media platform — Electron + React Native mobile sync, offline-first architecture

▸ Cross-Platform Deployed

Sovereign Bridge
Zero-cloud PC ↔ Mobile sync system — WebSocket tunneling, E2EE clipboard/file transfer

▸ Zero Cloud Dependencies

▸ WORK EXPERIENCE

Independent Software Developer
Jan 2026 — Present

VERITAS Ecosystem · Self-Employed · Bethalto, IL

- Designed and shipped 33 production applications in 12 weeks as a self-taught developer
- Built full-stack systems spanning desktop (Electron), web (React, Flask), mobile (React Native), and cloud (GCP)
- Authored 431 test specifications across the portfolio with zero production error rate
- Published research artifacts to Zenodo (DOI) and HuggingFace; contributed 21 test specs to Anthropic Cookbooks

Field Service Technician II
Oct 2024 — Feb 2026

Technical Safety Services · St. Louis, MO

- Certified Class II BSCs, Laminar Flow Hoods, Chemical Fume Hoods, and Isolators per NSF/ANSI 49, ISO 14644, USP ⟨797⟩/⟨800⟩
- Executed HEPA/ULPA filter integrity testing, particle counting, pressurization monitoring, and airflow visualization
- Generated audit-ready certification reports meeting GMP and GDP standards
- Operated in BSL-2 and BSL-3 environments under strict biohazard and chemical protocols

Quality Control Inspector
Jul 2023 — Mar 2024

Blount Fine Foods · Fall River, MA

- Led quality audits and inline adjustments to meet production specifications; investigated and corrected customer complaints
- Coordinated specs and cooking instruction changes for Right First Time initiatives with cross-functional teams
- HACCP trained; enforced allergen control programs, GMPs, and SOPs across production floor

Thermal Spray Technician
Feb 2021 — Jun 2023

HTS Coatings · Madison, IL

- Programmed and operated manual, semi-automatic, and robotic spray systems; created spray recipes and sequences
- Inspected surfaces using micrometers, calipers, elcometers, and profilometers for compliance with specifications
- Cross-trained technicians and drove continuous improvement of cell processes

Delivery Driver — Home Medical Equipment
2024

Enos Home Medical · New Bedford, MA

- Delivered and installed medical equipment (beds, oxygen systems); responded to emergency calls on-call

Maintenance Technician
May 2020 — Feb 2021

IMC Outdoor Living · Godfrey, IL

- Preventative and emergency maintenance across industrial machinery; welding, fabrication, electrical, HVAC, plumbing

Restaurant Manager
Nov 2017 — Sep 2019

Pizza Hut · Godfrey, IL

- Full P&L ownership, staffing, inventory control, and customer satisfaction for store operations

TECHNICAL SKILLS

Python · JavaScript · Electron / Node.js · React Native · Flask · HTML / CSS · SQLite / PostgreSQL · Git / GitHub · REST APIs · WebSockets / IPC · GCP Cloud Run · Kotlin / Android · C++ / Emscripten · Webpack / Vite · Agile · AI Integration · Mathematical AI Bounding · Prompt Engineering · System Architecture · Root Cause Analysis · ISO 9001 · Technical Documentation

CERTIFICATIONS

✓ Forklift Certification ✓ CPR Certified ✓ ServSafe Food Manager ✓ Non-CDL Class C ✓ Electric Pallet Jack ✓ Driver's License

EDUCATION

Taunton High School · Taunton, MA · Graduated 2008
Self-directed technical education: full-stack development, AI systems, cryptographic verification